Characteristic torsion changes the field and pairing conditions

On the actual nonsingular F2curve E:y²+xy=x³+1, Lean proves that every geometric2-torsion point is rational, yet2does not divide2^1−1. This refutes the unrestricted set-inclusion version of the torsion-field criterion. It also proves that every μ2-valued map on this nontrivial geometric torsion group fails pointwise left nondegeneracy. [pcc:unverified]

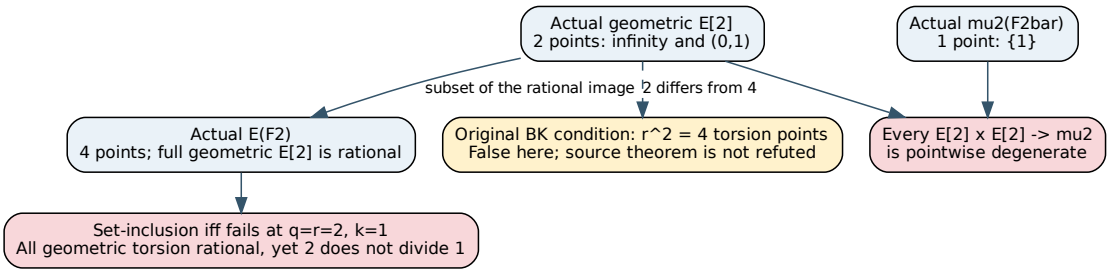

Figure 1: Actual Mathlib groups and algebraic closure. Counts are kernel-proved cardinalities, not measurements. The r²point-count condition is stronger than set inclusion at characteristic torsion. [pcc:unverified]

Object	Checked cardinality	Actual meaning
$E(\mathbb{F}_2)$	4	Nonsingular rational points, including infinity
$E[2](\overline{\mathbb{F}}_2)$	2	Only infinity and (0,1), both rational
$\mu_2(\overline{\mathbb{F}}_2)$	1	Every square root of 1 equals 1

BK1998 Theorem1 uses the existence of r²torsion points, not merely containment of the geometric torsion set. Its proof first obtains r not dividing q from divisibility. Atq=r=2the geometric torsion has2points, not4, so the primary source is not refuted. The compressed set-inclusion version needs a prime-to-characteristic/rank-two qualification. The actual corpus node already includes gcd(r,q)=1; this is its excluded-case negative control, not a disproof of that node. The earlier F7/r3counterexample separately preserves the r not dividing q−1 exclusion. [pcc:unverified]

For the pairing obstruction, the proof identifies a nonzero actual point killed by2and a singleton root-of-unity value type. Every map has value1 on its entire row. This is stronger than checking one arbitrarily chosen bilinear map. It is explicitly a statement about geometric points; the finite-flat group-scheme Weil duality statement is not refuted, and actual Weil/Tate constructions on their qualified domains remain open. [pcc:unverified]

Source: BK1998, pp142–143, Conditions1–2/Theorem1/proof, DOI10.1007/s001459900040; PDF SHA256prefix ac837b8edc47. Full hashes/read scope: ../../CHARACTERISTIC_TORSION.md. Lean source: ../../lean/CharacteristicTorsionBoundary.lean, combined theorem characteristic_boundary_certificate. The full project gate passes 134 declarations, 0 failing and 0 trusted-native exceptions. Packet PCC-20261109-92d64d remains unsigned. Full transitive gate and failed diagnostics are retained under ../../evidence/characteristic-torsion-*. No new native population or frozen-data change; zero paid spend. [pcc:unverified]